

Threshold Entry in a Fixed-Prize Contest: Proofs

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Abstract

This note collects the proofs underlying a rebuilt model of threshold entry in a fixed-prize contest, developed after both referee reports on a prior submission identified the equilibrium analysis as unsound (Section 1). Distribution-free throughout: no distributional family is assumed except where explicitly flagged. Every claim is tagged **PROVED**, VERIFIED (numerically, not proved), or *REFUTED*, and Table 3 links each claim to the file and object that establishes it. Two refuted claims are the author's own conjectures; one is a referee's. Table 3 additionally records an *evidence tier* for each claim — **L** machine-checked in Lean 4, **S** exact symbolic derivation, **N** numerical sampling — because these are not equivalent grades of assurance and no claim here is machine-checked end to end from primitives (Section 2).

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1 Scope and verification methodology

Verification is split across three tools with an explicit division of labour. **SymPy** verifies the analytic content: exact symbolic integration and algebraic identities, checked against closed-form function families where the integrals evaluate exactly. **Lean 4 (core only, no sorry)** verifies the discrete and order-theoretic content: given the analytic sign of a step difference as an explicit hypothesis, the equilibrium structure (existence, uniqueness, the prefix property) is machine-checked independently of the analytic step. **SciPy/NumPy** supplies numerical evidence where neither symbolic tool applies: adversarial counterexample search, sampled comparative statics, and convergence diagnostics.

Mathlib is deliberately not used, so no measure-theoretic content is formalised in Lean directly; the analytic step enters the Lean development only as an explicit hypothesis, so the *logical dependency* of the equilibrium result on the analytic identity is still machine-checked, even though the identity itself is verified in SymPy rather than Lean. The reason for staying on core Lean is auditability, not availability — §2 states it, and `verify.sh` measures Mathlib’s presence on every run rather than asserting it either way.

All source files are included in the accompanying bundle. Run `./verify.sh` to reproduce every claim in Table 3; it regenerates `VERIFICATION.md` from the actual run, including an axiom audit covering every theorem declared in the Lean file, so that status file cannot drift from the suites it reports. This document is the canonical statement of the results; `VERIFICATION.md` records only what the checks printed.

A second and separate suite, `lit/verify_lit.sh`, covers the *positioning* claims — the assertions this document makes about what other papers prove and how their results relate to these. It is deliberately not folded into `./verify.sh`, because a failure there would mean a misreading of the literature rather than a defect in a result. Two positioning statements below rest on machine-checked artifacts in that suite: the dispersive-order bridge of §6.11 and the kernel correspondence of §6.13. Those artifacts verify *our reading* of the cited sources; they do not reprove anything in those sources.

2 What the three evidence tiers mean

The verification suite has three legs and they do not carry equal weight. Table 3 records which apply to each claim.

L — Lean 4. Kernel-verified. `lean/EntryContest.lean` compiles with no `sorry` and no user-declared axioms; `verify.sh` runs `#print axioms` on every declared theorem and fails if any is unaudited. Of the 52 theorems, 36 depend on no axioms at all and 16 depend only on `propext` and `Quot.sound`, both Lean core

axioms, pulled in by omega on Int arithmetic. These three counts are cross-checked against the generated VERIFICATION.md by checks/verify_docs.py, so they cannot drift from the audit as the development grows. Verified on Lean 4.33.1, most recently on 2 September 2026; the file also compiled unchanged on 4.14.0 in an earlier session, which has not been retested since.

Two limitations are material and are stated here rather than left implicit. First, Mathlib is not used, so the development is deliberately confined to the *discrete and order-theoretic* content; every analytic step enters as an explicit hypothesis. The word *deliberately* is load-bearing and was previously doing work it had not earned: the bundle’s generated environment note used to record that Mathlib was unreachable, which was false on at least one machine the suite is run on. verify.sh now measures this on every run rather than asserting it, and VERIFICATION.md carries the finding. The reason for staying on core Lean is not availability but auditability — it is what allows #print axioms to certify every one of the theorems, which a Mathlib-based development could not do at the same granularity. In particular the antitonicity of Δ — the engine of P5 — enters as the hypothesis step_nonpos and is discharged not in Lean but at tier S. Second, the Lean theorems are conditional deductions: they establish that the stated conclusions follow from the stated hypotheses, not that the hypotheses hold for the model’s primitives. Both facts are by design, and together they mean that *no claim in this document is machine-checked from primitives to conclusion*. What is machine-checked is that the logical skeleton is sound given the analytic inputs.

S — SymPy. Exact symbolic computation, not machine-checked proof. checks/verify_sympy.py derives the difference identity (S1–S6), the P1 representation (S7), the sign of $\int \phi^2 dK$ (S8), the $\mu \rightarrow 0$ benchmark (S9), the P-MU difference identity and weight shape (S10–S12) and the two algebraic steps of P7 (S18–S20). These are exact rational-arithmetic results rather than floating-point samples, but they are produced by a computer algebra system whose correctness is assumed, not verified. Most are checked over stated families and are therefore instantiations of a universally quantified claim, not proofs of it; the exceptions are S1, S18 and S20, which are abstract in F, G, C or symbolic in m .

N — numerical. Sampling over finitely many economies. Strong evidence against a claim (a single violation refutes) and weak evidence for one. Where a claim rests on N alone this is flagged in the table, because such a claim is supported but not established.

The remaining analytic inputs. Three analytic facts are supplied rather than proved. Two enter as explicit hypotheses: the sign of the step difference (step_nonpos, discharged at S by S1–S6), and the divergence of κ at the support floor (discharged at S by S16). Both are stated as hypotheses in the Lean development and as assumptions in Section 4, so the dependency is explicit rather than hidden.

It is worth being precise about what S1–S6 does and does not leave family-dependent, because “discharged at tier S” understates it. The links carrying the argument are abstract in F, G, C and K : the algebraic step $H_{m+1} - H_m = -K\phi$ (S1), the chain rule (S2), the sign of the integrand (S3), and — the one that encodes an actual modelling assumption — the vanishing of the boundary term when $\phi(a) = \phi(b) = 0$ (S5). Of the two remaining, S4 is integration by parts, a theorem of calculus rather than a fact about this model, checked on five concrete pairs because SymPy cannot decide it in abstract-function form; and S6 instantiates the assembled identity on four closed-form families. So the family-dependence sits in the assembly and in a textbook step, not in the links that do the economic work.

The third is P7’s uniform bound. Its two algebraic steps are now verified at S: the factorisation $F^m - H_m = F^m(1 - G^{Q-1-m}C)$ holds for abstract F, G, C with no family assumed (S18), the bracket is non-negative because $G^e \leq 1$ on $[0, 1]$ and $C \leq 1$ (S19), and the base integral $\int_0^1 u^m du = 1/(m+1)$ is exact *symbolically* in m rather than over finitely many exponents (S20). What remains assumed is monotonicity of the integral functional — the passage from $H_m \leq F^m$ pointwise to $\int H_m dF \leq \int F^m dF$, and from $H_m \geq 0$ to $\int H_m dG \geq 0$.

That is the same functional-monotonicity hypothesis already supplied explicitly in the Lean development for P2 (§6.2), so P7 introduces no new analytic assumption; it reuses one that was already on the books.

Why P7 is not checked end to end. The conclusion $\Delta(m) \leq V/(m+1)$ could be evaluated exactly over the admissible power families and compared against the cap. That test was tried and rejected as uninformative: over those families $\max(m+1)\Delta(m) \approx 0.46$, and ≈ 0.79 over a widened set, so the strictly tighter caps $1/(m+2)$, $1/(m+3)$ and $1/(m+4)$ pass on every case as well. A check that cannot separate the true bound from a tighter false one is not evidence for the true bound, so the *proof steps* are verified rather than the conclusion. The numerical block in `checks/verify_p679.py` is retained as corroboration and is subject to the same caveat.

A caveat on P-MU. The hump of the weight does not by itself show that first-order stochastic dominance leaves the sign of the P-MU comparison open. Two exact witnesses with $F \leq G$ and opposite signs show it, and both are proved in Lean (§6.13). An earlier version of this paragraph rested the point on S11 and S12, which do not establish it.

Why row BM is in the table at all. Every other row records a result derived *within* the model. Row BM does not: it is a statement about utility functions, upstream of the model entirely, and assumptions do not otherwise earn rows — there is none for $V > 0$, none for independence of the draws from wealth, none for common knowledge of the wealth profile, though that last is load-bearing. The closest precedent, $\kappa \rightarrow \infty$ as $w \downarrow c$, is likewise an assumption discharged at tier S (S16) and has no row of its own.

BM is listed because it is not the assumption. The assumption is (2), which sits in §4 with the others. What is listed is the claim that the assumption is *strictly* weaker than the concavity it replaces, and that claim does real work: without it the §7 sentence asserting a weaker hypothesis is a rewording rather than a weakening. It also rests on a single knife-edge construction, so it is exactly the kind of support the tier column exists to make visible. The implication in the other direction, concavity $\Rightarrow$ (2), is elementary and is left in the prose, matching how $\kappa \rightarrow \infty$ is handled.

3 Verification index

Claim	Statement (short)	Status	Tier	Verified in
BM §1	Burden-monotonicity is <i>strictly</i> weaker than concavity: a non-concave u satisfies it	PROVED	S	<code>checks/verify_sympy.py</code> (S22a oscillation identity, abstract in k, c, w ; S22b amplitude exactly zero when $p \mid c$; S22c reduced burden strictly decreasing; S22d $u'' > 0$ exhibited). Control: S23, $p = 0.7$ breaks (2) at $w = 1.3$. The implication concavity $\Rightarrow$ (2) is one line and is proved in §4; S21 records the identity and instantiates it. This row is about the primitives, not a result derived within the model — see §2.

Claim	Statement (short)	Status	Tier	Verified in
ANON §4	Anonymity is delivered by wealth-independent draws, and count invariance does <i>not</i> extend beyond it	REFUTED	L	lean/mathlib/Anonymity.lean (gain_anonymous proves §8 on the induced primitives; anon_witness_zero, anon_witness_positive, an exact two-size witness with ability falling in wealth, at $\mu = 0$ and for every small $\mu > 0$). For ability rising with wealth, checks/verify_anonymity.py A1-4b reports 0 violations in 20,000 economies, which is support, <i>not</i> proof (§2, tier N for a universal claim). Bounds the scope of every result in §6.
P1 §6.1	$\Delta(m) = V \mathbb{E}[\varphi(M_m)]$	PROVED	S	checks/verify_sympy.py (S7, exact); checks/run_all.py (P1, max err 3×10^{-6})
P2 §6.2	FOSD: $F \leq G$ pointwise	PROVED	L	lean/EntryContest.lean (below_subset_of_pathwise, cdf_le_of_pathwise); checks/verify_sympy.py (S13 pathwise gap, S14 exact CDF ordering); checks/run_all.py (P2, 0 violations, $4\mu \times 3$ families)
P3 §6.3	$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \varphi^2 dK$	PROVED	S	checks/verify_sympy.py (S1–S6, exact); checks/run_all.py (P3, 4th-order convergence to 8×10^{-12})
P3-cor §6.3.1	Δ strictly antitone in m	PROVED	L	lean/EntryContest.lean (antitone_of_step, strict_of_step); checks/run_all.py (P4, 1500 adversarial pairs, 0 violations)
P5 §6.4	Existence & uniqueness of k^* (prefix property)	PROVED	L	lean/EntryContest.lean (3 theorems, listed in §6.4); checks/run_all.py (P5)
P5-inv §6.4.1	Every pure-strategy equilibrium has count k^* ; identities pinned under a stated condition	PROVED	L	lean/EntryContest.lean (count_le_kstar, count_ge_kstar, equilibrium_count_unique, members_below_kstar); lean/mathlib/MeasurementMap.lean (count_invariance, with the counting facts proved); checks/verify_equilibria.py (E1–E6: 20,000 instances, corroboration)

Claim	Statement (short)	Status	Tier	Verified in
N4 §6	Assortative set is always an equilibrium and minimises total κ ; ranking is an unweighted utility sum, not resource efficiency	PROVED	L	lean/EntryContest.lean (cost_exchange_le, totalCost_cons, surplus_gap_is_cost_gap); checks/verify_equilibria.py (N4, 6000 economies, 1335 with multiplicity; control 3892; N4-WELFARE, 1344 multiplicity economies)
P6 §6.5	$\mu \rightarrow 0 \Rightarrow \Delta(m) = 1/(m+2)$, indep. of Q	PROVED	S	checks/verify_p679.py (P6 block); checks/verify_sympy.py (S9); checks/run_all.py (P6)
R1 §6.6	Collapse threshold $\bar{Q}$ with $k^* = 0$	REFUTED (own)	L	lean/mathlib/Refutations.lean (bern_Delta_lower, bern_Delta_limit, r1_lower_bound); checks/run_all.py (R1, illustration)
P7 §6.7	$\Delta(m) \leq V/(m+1)$; $k^* \leq V/\kappa$, unif. in Q	PROVED	S/L	Algebraic steps S: checks/verify_sympy.py (S18 dominance factorisation, abstract F, G, C ; S19 $G^e \leq 1$ on $[0, 1]$; S20 $\int_0^1 u^m du = 1/(m+1)$, symbolic in m). Deduction bound $\Rightarrow$ cap L: lean/EntryContest.lean (entry_index_bounded). Integral monotonicity assumed, see §2. Corroborated N : checks/verify_p679.py, checks/run_all.py.
R2 §6.8	k^* decreasing in Q for all $\mu > 0$	REFUTED (referee)	L	lean/mathlib/Refutations.lean (bern_Delta_strictMono, Delta_mono_uniform); lean/mathlib/FallWitness.lean (fall_from_one, rise_then_fall); checks/run_all.py (R2, illustration)
P8 §6.9	k^* nondecr. in V , nonincr. in c	PROVED	L	lean/EntryContest.lean (entry_monotone, nonentry_monotone); checks/verify_p89.py (P8a, P8b)
P-UNIQ §1	Single crossing $\Rightarrow k^*$ unique; Δ invariant to wealth	PROVED	L	lean/EntryContest.lean (single_crossing); checks/verify_p89.py (P-UNIQ, 300 economies)

Claim	Statement (short)	Status	Tier	Verified in
P9 §6.10	MPS sign flips exactly at $\bar{w} = \bar{\mu}$ (full effect)	PROVED	L	lean/EntryContest.lean (linSpread_isPivotSpread, reducing P9 to P9-gen; linSpread_monotone); checks/verify_sympy.py (S15a- c); checks/verify_p679.py (P9 block); checks/verify_p89.py (both branches: 230 + 4000 cases)
P9-gen §6.11	Sign rule for arbitrary pivot- spreads (pivot x_0 , not the mean)	PROVED	L	lean/EntryContest.lean (cost_falls_above_pivot, cost_rises_below_pivot, order_preserved; composed into entry_preserved_above_pivot, nonentry_preserved_below_pivot, entry_set_grows_above_pivot); checks/verify_p9gen.py (4 fam- ilies, 0 violations; 729 control violations)
R1 §10	Margin condition on two sorted profiles; the marginal entrant's rank settles the rise branch and the first outsider's rank the fall branch; the rank-by-rank form and pivot-spreads are special cases; the mean cannot replace the pivot	PROVED	L	lean/mathlib/Spreads.lean (margin_rise, margin_fall, tail_rise, tail_fall, pivot_rise, pivot_fall, mps_lowers_count); lean/EntryContest.lean (entry_preserved_of_up_to_margin, nonentry_preserved_of_beyond_margin, rank-by-rank form); checks/verify_p9gen.py (R1, R1-SCOPE, R1-WEAK, R1-FREE, illustration)
BAND §6.11.3	The margin condition is silent only on the band $d_{k^*} < 0 < d_{k^*+1}$; dispersive changes never reach the band; inside the band the dis- placement signs determine noth- ing, for mean-preserving spreads too	PROVED	L	lean/mathlib/Spreads.lean (branch_of_not_band, not_band_of_single_crossing, pivot_single_crossing, mps_band_signs, mps_witness_counts, mps_band_both_directions, mps_band_both_directions_positive, band_both_directions, band_both_directions_positive); lean/EntryContest.lean (band_empty_iff, band_nonempty_of_lt, superseded L/M band); checks/verify_tailband.py (T1–T5, superseded band, illustration). Resolves open item 3.

Claim	Statement (short)	Status	Tier	Verified in
P9-str §6.12	Strong spread $\Rightarrow k^*$ falls strictly	PROVED	L	lean/EntryContest.lean (linSpread_strict_anti_in_lam, linSpread_unbounded_below, entry_ceiling_of_marginal_exit, strict_drop_of_marginal_exit); checks/verify_sympy.py (S16 $\kappa \rightarrow \infty$, S17 strict monotonicity); checks/verify_p9gen.py (P9- STRICT, 400 economies)
P-MU §6.13	Exact condition for sign of $d\Delta(0)/dQ$; $\Delta(0, \cdot)$ quasi- concave in Q when φ is single-peaked	PROVED	L	lean/mathlib/PMU.lean (pmu_identity, pmu_sign_iff, kernel_rises, kernel_falls, pmu_single_crossing, pmu_quasiconcave, pmu_quasiconcave_uniform); lean/mathlib/PMUWitness.lean (fbsd_does_not_sign); checks/verify_sympy.py (S10– S12); checks/verify_p89.py (P-MU, observed sign)
Alg	$KF - KG = K(F - G)$ (factorisa- tion used in P3)	PROVED	L	lean/EntryContest.lean (step_factor, step_sign)

4 Primitives

One prize of value $V > 0$, *exogenous and fixed* — decoupled from the wealth distribution. This decouples the prize from inequality, which is the fix for the prize/affordability conflation raised in referee comments (Section 8.2).

The prize is allocated by rank rather than purchased, and that is a modelled feature rather than a convenience. Cole et al. [1992] interpret status as “a ranking device that determines how well he or she fares in the nonmarket sector”, and Cole et al. [1995] observe that when desirable goods “are allocated as prizes rather than sold, the standard welfare theorems regarding the Pareto optimality of the outcomes no longer apply” — a remark in their concluding discussion rather than a theorem, but the right characterisation of the object. Nothing below inherits their machinery: by P1 the gain is proportional to V , so V enters this model only as a scale factor, and all that is used of it is that it is fixed and rank-allocated.

Q challengers plus one incumbent. Each player chooses $e_i \in \{0, 1\}$: pay nominal cost c to acquire social capital. After the entry decision, draws r_i, s_i independent across players *and independent of the wealth profile*, from *arbitrary* distributions, $s_i \geq 0$. Score

$$\sigma_i = \mu r_i + (1 - \mu) s_i e_i.$$

Highest score takes the prize. Static, one shot.

Independence of the draws from wealth is a substantive modelling commitment and is where Proposition 8 bites: it is what makes F and G , and hence Δ , functions of the entrant *count* alone rather than of which wealths entered. Economies in which wealth buys ability — so that s_i is drawn conditionally on w_i — are outside the scope of everything below, and deliberately so: the model isolates the pure affordability channel.

What lies outside that scope, and why it matters. The exclusion is not a formality, and the boundary is demonstrated rather than asserted. Let challenger i draw s_i with $\Pr(s_i \leq x) = x^{n_i}$ on $[0, 1]$, the largest of n_i uniform draws, so that n_i is an ability and equal abilities reproduce the assumption above. Challenger i wins with probability $\int C \prod_{j \neq i} F_j dF_i$, where F_j is the score CDF of challenger j (`lean/mathlib/Anonymity.lean`, `winProb`). Three facts follow.

- With equal abilities, the gain to challenger i from entering against the set T of other entrants is $\Delta(|T|)$, a function of the number of other entrants alone (`gain_anonymous`). This is Proposition 8 proved on the model's induced primitives.
- For ability falling with wealth, count invariance is **false**. Take three challengers with wealth $(11/4, 9/4, 2)$ and abilities $(1, 2, 3)$, CRRA $\gamma = 2$, $c = 1$, $V = 1$, and an incumbent of ability 1 who invests. At $\mu = 0$ the gain to challenger i against T is $n_i/(1 + n_i + \sum_{j \in T} n_j)$ (`gain_zero`), the costs are $\kappa = (16/77, 16/45, 1/2)$, and the *poorest* challenger entering alone and the two *richest* entering together are both equilibria, with counts 1 and 2 (`anon_witness_zero`). Every equilibrium condition holds strictly and every win probability is continuous in μ (`winProb_tendsto`), so both equilibria persist for every small $\mu > 0$, where every score law has no atoms and the primitives hold (`anon_witness_positive`).
- For ability rising with wealth, count invariance held in every economy sampled, with $\Pr(s_i \leq x) = x^{a_i}$ and $a_i = e^{\theta z_i}$, $\theta > 0$ (`checks/verify_anonymity.py`, A1-4b). This is *support and not proof*. Zero violations in a finite sample is tier N evidence for a universal claim, in the sense of §2.

An earlier version of this paragraph rested the second fact on a witness with $Q = 4$, $\mu = 1/2$ and $\theta = -3/2$ computed by quadrature (`checks/verify_anonymity.py`, A1-5), which remains there as an illustration.

The mechanism of the failure is worth stating, because it identifies what anonymity is really buying. Under a negative tilt the ablest challenger is also the one facing the highest entry cost, so the question of *how many* enter and the question of *which* of them enter cease to be separable — and it is precisely their separation that Proposition 6.4.1 turns into a count that every equilibrium shares. Since every comparative static below is a statement about that count, the assumption is load-bearing for the whole of Section 6, not only for Proposition 8. Positive correlation between wealth and ability appears to be harmless because it reinforces the cost ordering rather than opposing it; that direction is not claimed.

Payoffs. Writing u for the utility of wealth,

$$U_i = u(w_i - c e_i) + V \cdot \mathbf{1}\{i \text{ wins}\}, \quad (1)$$

with u increasing and satisfying the burden condition stated below. Two features of (1) do real work and are worth stating rather than leaving to be inferred from the entry condition. The cost is paid *inside* u , so it is felt in utility units and its burden falls with wealth; the prize is *additively separable* and enters linearly, so it is valued identically by rich and poor and carries no wealth effect of its own. Inequality therefore operates through exactly one channel, the cost side.

This is the *privilege* contest of Schroyen and Treich [2016] rather than their rent-seeking contest, and the difference is not cosmetic: were V money added to the resource, it would sit inside u , the two wealth effects would oppose, and under CARA they would exactly cancel. Keeping V outside u is licensed by what the prize is — a rank-allocated non-market good in the sense of Cole et al. [1992], not a transfer that can be spent.

Induced score CDFs: G for a non-investor (μr), F for an investor ($\mu r + (1 - \mu)s$). All results are stated in F and G only. Write

$$\varphi \equiv G - F \geq 0.$$

Regularity: F, G continuous, common lower endpoint, ϕ vanishing at both endpoints of the union of supports.

Wealth heterogeneity among challengers, $w_i \sim \Lambda$, with support bounded below by c and $\kappa(w) \rightarrow \infty$ as $w \downarrow c$. Utility cost of entry $\kappa(w) = u(w) - u(w - c)$, continuous — the only channel through which inequality operates.

Burden-monotonicity. The assumption on u is

$$w' > w \implies u(w') - u(w' - c) < u(w) - u(w - c), \quad (2)$$

that is, κ is strictly decreasing in w . Note the scale at which this is stated: it constrains the increments of u over spans of the *fixed* length c , not the curvature of u at a point.

Strict concavity of u is sufficient, and this is how earlier versions of this document stated the primitive. The implication is immediate from

$$\kappa'(w) = u'(w) - u'(w - c),$$

which is negative whenever u' is strictly decreasing (S21). It is *strictly* sufficient: there are non-concave u satisfying (2), and one is exhibited in Claim 1 below. Nothing in this document uses concavity anywhere else. Every proof that touches the cost side runs on κ , and the only other appearance of u is $u'(w - c) > 0$ in Proposition 6.9, which is monotonicity rather than curvature.

The Lean development was always at this generality: `lean/EntryContest.lean` takes burden-monotonicity as the hypothesis `hkap :  $\forall ab, a \leq b \rightarrow \kappa(b) \leq \kappa(a)$` directly, and the symbol u does not occur in the file at all. So the restatement above closes a gap between the prose and the formal layer rather than changing what was proved. The prose had claimed *less* than had been established.

Claim 1 (Burden-monotonicity is strictly weaker than concavity). *There exists an increasing u , non-concave on every neighbourhood of a set of points of positive measure, satisfying (2) on a non-degenerate interval. Hence the class of admissible u is strictly larger than the concave class.*

Proof. Take $u(x) = \sqrt{x} + \varepsilon \sin(kx)$ with $c = 1$, $\varepsilon = 3/50$ and $k = 2\pi/p$. For any k, c and w ,

$$\sin(kw) - \sin(k(w - c)) = 2 \sin\left(\frac{kc}{2}\right) \cos\left(k\left(w - \frac{c}{2}\right)\right),$$

verified symbolically in k, c, w with no family assumed (S22a). The burden therefore inherits the oscillation only through the amplitude $2\varepsilon \sin(kc/2)$, which vanishes exactly when kc is a multiple of 2π , i.e. when the period p divides c . At $p \in \{1, \frac{1}{2}, \frac{1}{3}\}$ the amplitude is exactly zero (S22b), so $\kappa(w) = \sqrt{w} - \sqrt{w - 1}$ identically, which is strictly decreasing (S22c). Yet $u''(x) = -\frac{1}{4}x^{-3/2} - \varepsilon k^2 \sin(kx) > 0$ at every point with $\sin(kx) = -1$ in the range, and such points are exhibited (S22d). $\square$

How much room this buys is open, and the example is knife-edge. The amplitude $2\varepsilon \sin(kc/2)$ vanishes only on a measure-zero set of periods. At $p = 0.7$, which does not divide c , (2) fails outright, and the violation is exhibited at $w = 1.3$ (S23). Claim 1 establishes that the containment is *proper*; it does not establish that the additional room is economically substantial, and it should not be read as evidence that non-concavity is generally harmless. Characterising the tolerance — the conjecture being that it is governed by the width of the convex region relative to c , since a burden reading only wealth differences of size c cannot see a convex stretch much narrower than c — is open and is recorded as such in `TODO.md`.

Information. The realised wealth profile $(w_1, \dots, w_Q)$ is *common knowledge* among the challengers, and entry decisions are simultaneous. This assumption is load-bearing and was left implicit in earlier versions of this document. It is what licenses writing the entry condition on the order statistics $w_{(1)} \leq \dots \leq w_{(Q)}$ as in (3), and hence what makes k^* a *deterministic* count — one that is the same in every pure-strategy equilibrium, by Proposition 6.4.1 below.

It does *not* pin the identities of the entrants, and an earlier version of this document claimed that it did. It does not: with $\Delta = (12, 10, 1)$ and $\kappa = (2, 5, 8)$ — admissible, since Δ is strictly antitone and κ increasing in the wealth ranking — the entrant sets $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$ are all equilibria, the last of these with the *richest* challenger absent, because joining as a third entrant is unprofitable even for her while the two poorer entrants are each content facing one rival. What complete information delivers is the count and the existence of the assortative equilibrium, not uniqueness of the entrant set.

The alternative — privately known entry costs — yields a symmetric Bayesian threshold in cost space and a *binomial* entrant count, as in Moreno and Wooders [2011]; there the number of entrants and the identity of the entering set are both random, being determined by unobserved cost draws. The mechanical difference is sharper than the informational one: their equilibrium threshold is a single number t , against which every buyer compares her own cost, because under private information all buyers face the same expected utility. Here the comparison value depends on *rank* — the j -th richest faces $\Delta(j-1)$ — because she knows how many richer rivals enter ahead of her. A rank-dependent rule collapses to a flat one only if Δ is constant, which Corollary 6.3.1 forbids. The two models are genuinely different, and the difference is exactly this assumption, not the shape of κ . The complete-information reading is also what the verification artefacts implement: both `lean/EntryContest.lean` and the Python suites operate on an explicitly ordered wealth profile.

Under complete information with heterogeneous $\kappa(w_i)$ there is no symmetry to break, so no mixing is required and no tie-breaking rule is needed. This is the respect in which the present model differs from the endogenous-entry literature with identical agents, where the entrant count is pinned but the identities are not — resolved there by mixed entry [Levin and Smith, 1994], by sequential arrival [Fu and Lu, 2010], or by private random delays [Morgan et al., 2012].

Opponent maximum when m other challengers invest: $H_m = F^m G^{Q-1-m} C$, where $C \in \{F, G\}$ is the incumbent’s CDF. Gain from investing:

$$\Delta(m) = V \left[\int H_m dF - \int H_m dG \right].$$

5 Terminology

Every English term used in a claim or proof below is bound here to a symbol built from the primitives of Section 4. Terms are listed only where they carry load in a stated result; where a symbol alone would do, no term is introduced. Nothing in the table is a new object — each entry is a name for something already defined — so the table can be read as a translation key rather than as a second set of definitions.

Term	Symbol	Definition
<i>Players and actions</i>		
challenger	$i \in \{1, \dots, Q\}$	A wealth-heterogeneous player choosing whether to compete.
incumbent	$C \in \{F, G\}$	The additional player, entering exogenously; appears only through her score CDF C in H_m .
invest, enter	$e_i = 1$	Pay c and draw the two-component score.
entrant	i with $e_i = 1$	A challenger who invests.
outsider	i with $e_i = 0$	A challenger who does not.

Term	Symbol	Definition
entrant set	$S \subseteq \{1, \dots, Q\}$	$S = \{i : e_i = 1\}$.
prize	$V > 0$	Fixed, exogenous, rank-allocated; enters (1) additively, outside u .
<i>Wealth and cost</i>		
wealth	w_i	Challenger i 's resource, with $w_i > c$.
wealth profile	$(w_1, \dots, w_Q)$	The realised vector; common knowledge.
wealth distribution	Λ	The law from which the profile is drawn.
rank	j	Index into the descending order statistics $w_{(1)} \geq \dots \geq w_{(Q)}$; rank 1 is richest.
entry cost (utility)	$\kappa(w) = u(w) - u(w - c)$	The burden of c in utility units.
burden-monotonicity	κ strictly decreasing in w	(2): the assumption on u , stated at the scale of c rather than pointwise. Implied by strict concavity, strictly weaker than it (Claim 1). The hypothesis hkap in the Lean development.
entry cost (resource)	c	The nominal cost, identical for all agents. Distinct from κ ; the two are never interchangeable.
affordability channel	κ	The sole route by which the wealth distribution affects behaviour, given (1).
<i>Scores and gains</i>		
score	$\sigma_i = \mu r_i + (1 - \mu) s_i e_i$	What the prize is awarded on.
score CDFs	F, G	Law of σ for an entrant (F) and an outsider (G); $F \leq G$ pointwise by P2.
dominance gap	$\varphi = G - F \geq 0$	The wedge investment buys.
opponent maximum	$H_m = F^m G^{Q-1-m} C$	CDF of the best rival score when m other challengers invest.
gain from investing	$\Delta(m) = V \int H_m dF - V \int H_m dG$	The right-hand side of the entry condition; antitone in m by Corollary 6.3.1.
anonymity	Δ a function of m alone	Δ depends on the <i>number</i> of rival entrants, not on which agents they are (Proposition 8); requires draws independent of wealth.
<i>Equilibrium objects</i>		
entry condition	$\kappa(w_{(j)}) \leq \Delta(j - 1)$	(3): rank j enters when her utility cost is no greater than her gain.
single crossing	$\kappa(w_{(j)}) - \Delta(j - 1)$ increasing in j	Lemma 1; what makes the entry set a prefix.
entry count	k^*	The largest k with the entry condition holding at every rank $j \leq k$.
count invariance	$ S = k^*$ for all equilibria	Every pure-strategy equilibrium has the same size (Proposition 6.4.1).
identity multiplicity	$S \neq S'$, $ S = S' = k^*$	Distinct entrant sets are equilibria at the same count.
assortative set	$\{(1), \dots, (k^*)\}$	The k^* ranks with the lowest κ , i.e. the k^* richest.

Term	Symbol	Definition
marginal entrant	rank k^* , wealth $w_{(k^*)}$	The last rank satisfying the entry condition.
inframarginal entrant	rank $j < k^*$	An entrant richer than the marginal one.
outsider rank	rank $j > k^*$	A rank at which the entry condition fails.
<i>Comparative statics</i>		
spread	$w \mapsto w'$	Any map between wealth profiles; results below never assume it preserves the mean unless said.
mean-preserving spread	$\sum_j (w'_{(j)} - w_{(j)}) = 0$ plus second-order dominance	The economic case of interest, not a hypothesis of Proposition 10.
displacement	$w'_{(j)} - w_{(j)}$	The rank-by-rank difference between the two sorted profiles.
quantile coupling	$\Lambda_\beta^{-1}(r_j) - \Lambda_\alpha^{-1}(r_j)$	The displacement read as a difference of quantile functions; always available, since both profiles are sorted.
pivot	x_0	The wealth at which the displacement changes sign.
pivot-spread	T with $w \geq x_0 \Rightarrow Tw \geq w$ and $w \leq x_0 \Rightarrow Tw \leq w$	A spread expanding on both sides of a single pivot; the dispersive case.
margin condition	$w'_{(k^*)} \geq w_{(k^*)}$; or $w'_{(k^*+1)} \leq w_{(k^*+1)}$	The two-rank hypothesis of Proposition 10, read at the marginal entrant and the first outsider; implied by the pivot hypothesis and by the rank-by-rank condition of earlier drafts.
band	$d_{k^*} < 0 < d_{k^*+1}$	The margins on which the margin condition is silent; §6.11.3.
endogenous margin	k^* moving with the profile	The marginal entrant is a different agent before and after; Proposition 9.
<i>Aggregates</i>		
total entry cost	$\sum_{i \in S} \kappa(w_i)$	In utility units. Its resource counterpart is $ S c = k^*c$, identical across equilibria.
sum of payoffs	$\sum_i U_i$	Unweighted, hence an interpersonal comparison of the cardinal index u ; see the discussion after Proposition 6 for what it does and does not entail.
selection	a rule picking one equilibrium	Not a refinement: nothing claims players coordinate on it.

Three pairs are deliberately kept apart because conflating them is what earlier drafts did. *Entry cost* in utility (κ) versus in resources (c): only the first varies with wealth, and only the second is what society gives up. *Count invariance* versus *identity multiplicity*: the first holds, the second is real, and every comparative static below is about the count for that reason. *Spread* versus *margin condition*: the propositions are stated on the second, and the first is what they are applied to.

6 Proofs

6.1 P1: representation

Proposition 1. $\Delta(m) = V \mathbb{E}[\varphi(M_m)]$.

Proof. By parts, $\int H dF = 1 - \int F dH$ and $\int H dG = 1 - \int G dH$, the boundary terms vanishing since HF and HG run from 0 to 1. Subtracting, $\int H dF - \int H dG = \int (G - F) dH = \mathbb{E}[\varphi(M_m)]$. $\square$

Replaces the entire $p_1 < 1/(Q+1) < p'_1$ apparatus of the prior version with one distribution-free object.

6.2 P2: first-order stochastic dominance

Proposition 2. $F \leq G$ pointwise, strictly on an interval if s has positive mass on $(0, \infty)$.

Proof. $X = \mu r + (1 - \mu)s \geq \mu r = Y$ pathwise for $s \geq 0$, so for every threshold t the event $\{X \leq t\}$ is contained in $\{Y \leq t\}$, and monotonicity of probability under inclusion gives $F(t) \leq G(t)$. $\square$

The event inclusion — the whole probabilistic content of the step — is machine-checked as `below_subset_of_pathwise`, and its consequence for the CDFs as `cdf_le_of_pathwise`, with monotonicity of the probability functional supplied as an explicit hypothesis rather than proved (no measure theory is available in the Lean development; see §2). The pathwise gap $X - Y = (1 - \mu)s$ and the resulting CDF ordering are checked exactly in `checks/verify_sympy.py` (S13, S14). Hence $\varphi \geq 0$ and, by P1, $\Delta(m) \geq 0$: investing never hurts, with no assumption beyond $s \geq 0$.

6.3 P3: the difference identity

Theorem 1 (key result). For $K \equiv F^m G^{Q-2-m} C$,

$$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \varphi^2 dK \leq 0.$$

Proof.

$$\begin{aligned} \Delta(m+1) - \Delta(m) &= \int (H_{m+1} - H_m)(dF - dG) = \int K(F - G)(dF - dG) \\ &= \int K\varphi(dG - dF) = \int K\varphi d\varphi = \int K d\left(\frac{\varphi^2}{2}\right) \\ &= \left[\frac{K\varphi^2}{2}\right] - \frac{1}{2} \int \varphi^2 dK = -\frac{1}{2} \int \varphi^2 dK, \end{aligned}$$

the boundary term vanishing because $\varphi = 0$ at both endpoints. K is a product of CDFs, hence non-decreasing, so $dK \geq 0$. $\square$

6.3.1 Corollary: monotonicity

Corollary 1. Δ is non-increasing in m , strictly wherever $\varphi \neq 0$ on $\text{supp}(dK)$.

This refutes the author's own earlier conjecture. An initial sketch argued that because φ is hump-shaped (zero at both ends, positive between), $\mathbb{E}[\varphi(M_m)]$ need not fall when M_m rises stochastically, so monotonicity would need a hazard-rate or single-crossing condition. That reasoning was wrong: the integration by parts shows the hump shape is irrelevant, because φ enters *squared* against dK . Monotonicity is *unconditional*, and no auxiliary condition is needed.

6.4 P5: equilibrium existence and uniqueness

Proposition 3. *Let κ be the marginal entrant's utility cost. Since Δ is strictly decreasing (§6.3.1), the set $\{m : \Delta(m) \geq \kappa\}$ is a prefix $\{0, \dots, k^* - 1\}$, so*

$$k^* = \#\{m : \Delta(m) \geq \kappa\}$$

is the unique pure-strategy equilibrium number of investing challengers, with $k^ = 0$ and $k^* = Q$ as boundary cases.*

Proved in Lean as `enters_downward_closed`, `cutoff_unique`, and `equilibrium_is_threshold` (`lean/EntryContest.lean`), none depending on any axiom.

This delivers the referees' three-regime structure (full withdrawal / interior / full participation) as a *derived* result rather than an assumption, and is exactly the asymmetric k -of- Q structure requested in review (§8.1).

6.4.1 P5-inv: the count is an invariant of the game

Proposition 6.4 constructs the assortative equilibrium and shows the entry set is a prefix *of that schedule*. Since other equilibria exist (§4), the question is what survives across all of them. The answer is everything this document goes on to use.

Proposition 4 (count invariance). *Index challengers in increasing order of κ , equivalently decreasing wealth. Let S be any pure-strategy equilibrium entrant set: every $i \in S$ satisfies $\kappa_i \leq \Delta(|S| - 1)$ and every $i \notin S$ satisfies $\kappa_i > \Delta(|S|)$. Then $|S| = k^*$.*

Proof. Write $k = |S|$. Since S has k distinct indices, some member sits at index $k - 1$ or higher, and κ is increasing, so $\kappa_{(k)} \leq \Delta(k - 1)$; maximality of k^* gives $k \leq k^*$. Conversely, suppose $k < k^*$. Single crossing propagates the entry condition down from k^* , so $\kappa_{(k+1)} \leq \Delta(k)$. As $|S| = k$, at least one of the $k + 1$ cheapest agents lies outside S , and that agent has $\kappa \leq \Delta(k)$ — contradicting the outsider condition. Hence $k = k^*$. $\square$

The two combinatorial steps, some member at index $\geq k - 1$ and some index $\leq k$ omitted, are counting facts about a k -element set of indices. Both are proved (`lean/mathlib/MeasurementMap.lean`, `exists_mem_ge`, `exists_not_mem_le`) and composed with the order-theoretic core, `count_le_kstar` and `count_ge_kstar`, into `count_invariance`, which holds with no hypothesis beyond the model's. An earlier version left the two counting facts to enumeration in `checks/verify_equilibria.py`, which remains as corroboration on 20,000 random instances.

This is what licenses the rest of the document. P7, P8, P9, P9-gen, P9-str and P-MU are all statements about k^* , and k^* is exactly the quantity that does not vary across equilibria.

Proposition 5 (when identities are pinned). *If in addition $\kappa_{(k^*+1)} > \Delta(k^* - 1)$ — the first excluded challenger would decline even the most favourable slot — then every member of an equilibrium has index below k^* , so the assortative set $\{1, \dots, k^*\}$ is the unique equilibrium.*

Proof. A member satisfies $\kappa_i \leq \Delta(k^* - 1)$ by Proposition 4; if $i > k^*$ then $\kappa_i \geq \kappa_{(k^*+1)} > \Delta(k^* - 1)$, a contradiction. With k^* members all drawn from the first k^* indices, the set is exactly the prefix. $\square$

Machine-checked as `members_below_kstar`, and confirmed without exception on the 11,141 sampled instances satisfying the condition. Where it fails, assortativity remains a defensible *selection* — among count- k^* equilibria it is the one that fills the slots with the lowest-cost agents, and so minimises total realised entry cost — but that is an argument for a selection rather than a consequence of the primitives, and is treated as such here.

6.4.2 N4: assortative entry as a selection

Proposition 6.4.1 fixes the entrant count but not the entrant set: where $\kappa(w_{(k^*+1)}) \leq \Delta(k^* - 1)$, several distinct sets are equilibria. The assortative set — the k^* agents with the lowest entry cost — is therefore a *selection* and requires an argument rather than an assumption. The argument is that it is the cheapest, and that under anonymity cheapest means efficient.

Proposition 6 (assortative selection). *Among the pure-strategy equilibria of the entry game:*

1. *the assortative set $\{(1), \dots, (k^*)\}$ is always one of them;*
2. *it minimises the total entry cost $\sum_{i \in S} \kappa(w_i)$ actually paid; and*
3. *since Δ depends on the entrant count alone (Proposition 8) and that count is the same in every equilibrium (Proposition 6.4.1), the sum of the agents' payoffs differs across equilibria only through that total. The assortative equilibrium therefore maximises the unweighted sum of payoffs.*

Part (iii) is a statement about one particular aggregate, and the choice of that aggregate is a commitment, not a definition; the discussion following this proposition records what it does and does not entail.

Proof. (i) For $j \leq k^*$, single crossing gives $\kappa(w_{(j)}) \leq \kappa(w_{(k^*)}) \leq \Delta(k^* - 1)$, so every member is content; for $j > k^*$, $\kappa(w_{(j)}) \geq \kappa(w_{(k^*+1)}) > \Delta(k^*)$ by the definition of k^* , so no outsider wishes to join. (ii) All equilibria have k^* members by Proposition 6.4.1, and the k^* smallest values of a finite set minimise the sum over subsets of that size; the assortative set is exactly the k^* smallest κ . (iii) Fix two equilibria S, T with $|S| = |T| = k^*$. The prize term depends on $|S|$ only, so it cancels, leaving the difference in aggregate payoff equal to $\sum_{i \in T} \kappa(w_i) - \sum_{i \in S} \kappa(w_i)$. $\square$

The one-step exchange underlying (ii) — replacing a member by a cheaper non-member never raises the total — is machine-checked as `cost_exchange_le`, with `totalCost_cons` for additivity and `surplus_gap_is_cost_gap` for (iii), whose hypothesis $|S| = |T|$ is where anonymity enters. That the hypothesis is load-bearing and not decorative is checked directly: across 9022 pairs of equilibria the payoff gap equals the cost gap under a gain schedule that is a function of the count alone, and fails to in 3022 of the same pairs once the schedule is allowed to depend on *which* agents entered (`checks/verify_equilibria.py`, N4 and its control). The full minimisation in (ii) follows by iterating exchanges and is the standard sorting fact; it is checked numerically rather than proved in Lean. Over 6000 economies, of which 1335 exhibit genuine identity multiplicity, the assortative set is an equilibrium in every case and attains the minimum cost in every case; a non-assortative count- k^* set is *strictly* costlier in 3892 control cases, so the selection is not vacuous (`checks/verify_equilibria.py`, N4).

What the ranking in (iii) is, and what it is not. Three things have to be said, because (iii) is easy to over-read.

First, it carries no resource-efficiency content. Every equilibrium has k^* members and each pays the same access cost c , so the resources consumed are k^*c in all of them; the equilibria are indistinguishable in those terms. The ranking exists only because κ is a *utility* increment, $u(w) - u(w - c)$, which differs across agents of different wealth. Verified directly: over 1344 economies exhibiting genuine identity multiplicity, every equilibrium has the same size in every case, while the total κ differs strictly in every case (`checks/verify_equilibria.py`, N4-WELFARE).

Second, summing κ across agents is an interpersonal comparison of the model's cardinal utility index. That the index is cardinal is already assumed — the entry condition (3) compares a utility increment against a prize valued in the same units, which is the privilege-contest form of §4 — so no new cardinality is imported here. What is new is comparability across agents, and it is a normative commitment rather than a consequence

of the model. Its direction should be stated plainly: because κ falls in wealth, the unweighted sum is smallest when the wealthiest agents are the ones who enter, and the assortative set is exactly that set. In all 1344 multiplicity cases it is the equilibrium whose members are the richest, strictly (same check). The earlier reading of this as the choice of “a planner indifferent to identity” was wrong: a planner maximising the unweighted sum is not indifferent to identity, it strictly prefers wealthy entrants, and would rank the same allocation differently under any inequality-averse weighting.

Third, it does not compete with the welfare question the referee raised, and is not an answer to it. This proposition ranks equilibria *at* the equilibrium count k^* ; the welfare question is whether k^* is itself too large, since in a fixed-prize contest entry is rent-dissipating and the business-stealing logic of Levin and Smith [1994] applies. The two operate at different margins and are logically independent: (iii) selects among the sets the game admits, and says nothing about whether the game admits too many entrants. Nothing here is a welfare recommendation.

It remains a selection rather than a strategic refinement in any case: nothing says players coordinate on the cheapest equilibrium.

6.5 P6: the $\mu \rightarrow 0$ benchmark

Proposition 7. As $\mu \rightarrow 0$, $\Delta(m) = \frac{1}{m+2}$, independent of Q .

Proof. At $\mu = 0$, $Y = \mu r \equiv 0$, so G is the point mass at 0 and $G(x) = 1$ for all $x \geq 0$. Hence $G^{Q-1-m} \equiv 1$ and, with the incumbent investing, $H_m = F^m \cdot 1 \cdot F = F^{m+1}$. Then

$$\int H_m dF = \int_0^1 F^{m+1} dF = \frac{1}{m+2},$$

while $\int H_m dG = H_m(0) = F(0)^{m+1} = 0$ provided $s > 0$ almost surely. Therefore $\Delta(m) = 1/(m+2)$, with no dependence on Q . $\square$

Confirms Referee 2’s conjecture exactly. The Q -independence is structural: Q enters only through G^{Q-1-m} , which collapses to 1.

6.6 R1: no collapse threshold (refuted)

The author conjectured a finite $\bar{Q}$ beyond which $k^* = 0$, intended to replace the old mobility ceiling. **False.** Take r uniform on $[0, 1]$, an investment that fails ($s = 0$) with probability p and succeeds ($s = 1$) otherwise, $0 < \mu \leq \frac{1}{2}$ and an investing incumbent. Then for every $Q \geq 1$

$$\Delta(0, Q) = V \left(\frac{1-p^2}{2} - \frac{p(1-p)}{Q+1} \right) \geq V \frac{1-p}{2},$$

so a challenger whose cost is at most $V(1-p)/2$ enters for every Q , and $\Delta(0, Q) \rightarrow V(1-p^2)/2$ (lean/mathlib/Refutations.lean, bern_Delta, bern_Delta_lower, bern_Delta_limit). The mechanism is general. When non-investor scores never exceed x_G ,

$$\Delta(m, Q) \geq V \left(\int_{x \geq x_G} C F^m dF - \frac{1}{Q-m} \right) \quad (m < Q),$$

and the first term carries no Q (r1_lower_bound). An earlier version of this paragraph gave the limit as $\Pr(X > \mu) \cdot \Pr(\text{beat incumbent})$ on numerical evidence. That formula is withdrawn. In the family above $\Pr(X > \mu) = 1 - p$ (bern_investor_above) while the limit is $V(1-p)(1+p)/2$.

6.7 P7: saturation, uniformly in Q

Theorem 2. $\Delta(m) \leq V/(m+1)$ for every m and every Q ; hence

$$k^* \leq V/\kappa \quad \text{uniformly in } Q.$$

Proof. $H_m = F^m G^{Q-1-m} C \leq F^m$, since $G \leq 1$ and $C \leq 1$. So

$$\Delta(m) = \int H_m(dF - dG) \leq \int H_m dF \leq \int F^m dF = \frac{V}{m+1}.$$

Entry at index m requires $\Delta(m) \geq \kappa$, forcing $m+1 \leq V/\kappa$. The bound mentions no Q . $\square$

Saturation is not asymptotic but a *uniform explicit cap*, which replaces the prior version’s numerical claim. At $\mu \rightarrow 0$ the bound is nearly tight against P6’s $1/(m+2)$.

What is and is not new here. The accounting behind P7 — entrant count times entry cost cannot exceed prize mass — is not new, and the proposition should not be read as claiming a new inequality. Fu et al. [2015] obtain the same bound for identical agents with mixed entry: their rent-accounting inequality (2) states $[1 - (1-q)^M]V \geq Mq(\Delta + \mathbb{E}[x^\alpha])$, and every feasible point of their Definition 2 accordingly satisfies $Mq\Delta \leq V$, a cap on the *expected* entrant count. Fu and Lu [2010] reach the same relation as an equality, their (7) giving total effort as budget minus N times the entry cost.

Three things distinguish P7, and the contribution is exactly these. It bounds a *pure-strategy* count rather than an expected one, so the cap is a genuine finite index bound rather than a restriction on a probability. It is stated in *utility* units, κ rather than a nominal cost, which is what makes it interact with the wealth distribution at all. And it holds under *fixed rules with no designer*: Fu–Lu’s N is the count induced by an optimally chosen contest, whereas here the rules are given and the count is whatever the wealth ordering produces. Stripped of these, the shared content is a single monotonicity-and-transitivity step.

The verification splits along the two halves of the argument. The *implication* — if $(m+1)\Delta(m) \leq V$ and challenger m enters, then $(m+1)\kappa \leq V$ — is machine-checked as `entry_index_bounded` (depends on no axioms), though it is a single composition of monotonicity with transitivity and carries little content. The *premise* is where the work is, and its algebraic steps are exact (checks/verify_sympy.py): the dominance factorisation $F^m - H_m = F^m(1 - G^{Q-1-m}C)$ for abstract F, G, C (S18), the bound $G^e \leq 1$ on $[0, 1]$ that signs the bracket (S19), and $\int_0^1 u^m du = 1/(m+1)$ symbolically in m (S20). The one step not verified is monotonicity of the integral, which is assumed — see §2, where the reason the conclusion is *not* tested directly is also recorded. The quadrature block in checks/verify_p679.py (five values of μ , seven of Q up to 128, twelve of m , no violations, implied cap on k^* observed to hold) is corroboration rather than the basis of the claim.

6.8 R2: the Q comparative static (refuted)

Referee 2 conjectured that for $\mu > 0$, higher Q shrinks the pie among investors, so fewer invest. **Not general.** In the family of §6.6 with $0 < p < 1$, $\Delta(0, Q)$ rises strictly at every Q (`bern_Delta_strictMono`), so a challenger whose cost lies between $\Delta(0, Q)$ and $\Delta(0, Q+1)$ stays out with Q challengers and enters with $Q+1$. With r uniform the gain never falls in Q , for any law of $s \geq 0$, any $0 < \mu \leq 1$ and any incumbent (`Delta_mono_uniform`), because the weight of §6.13 vanishes above μ , where the non-investor’s score cannot reach, and below μ the investor’s law lies under the non-investor’s. The conjectured direction occurs too, for other primitives. Take r the smaller of two uniform draws, with density $2(1-r)$, an investment that fails with probability p and otherwise lifts the score by $1-\mu$, $\mu = 3/4$ and an investing incumbent. With $p = 1/4$ the gain falls from $Q = 1$ to $Q = 2$ (`lean/mathlib/FallWitness.lean`, `fall_from_one`), and with $p = 1/2$ the gain rises from $Q = 1$ to $Q = 2$ and falls from $Q = 2$ to $Q = 3$ (`rise_then_fall`). The sign

of the Q comparative static therefore depends on the primitives, and the law of r sets it. `checks/run_all.py` (R2) reports a crossover in μ by quadrature for r and s both Beta(2,2), which is an illustration and not evidence. An earlier version of this paragraph stated the crossover as a result.

6.9 P8: prize and affordability, separated

Sort challengers by wealth, descending. The j -th richest, entering as the j -th entrant, faces $j - 1$ others, so her entry condition is

$$\kappa(w_{(j)}) \leq \Delta(j - 1). \quad (3)$$

Lemma 1 (single crossing). *The left side of (3) is non-decreasing in j (κ is decreasing in w and $w_{(j)}$ is decreasing in j) and the right side is non-increasing in j (Corollary 6.3.1). Hence the entry set is a prefix and k^* is unique.*

This resolves what was previously listed as an open “fixed point” issue: no iteration is required. The reason is structural and worth stating separately.

Proposition 8 (anonymity). $\Delta(j - 1)$ depends on the number of entrants only, not on their wealth: the contest is anonymous, so F and G do not involve w . The wealth distribution therefore enters (3) only through $\kappa(w_{(j)})$.

Theorem 3 (P8). k^* is non-decreasing in V and non-increasing in c .

Proof. By P1, $\Delta(m) = V \mathbb{E}[\varphi(M_m)]$ is proportional to V , so raising V raises the right side of (3) pointwise; the entry set can only grow. For c : $\partial \kappa / \partial c = u'(w - c) > 0$, so raising c raises the left side pointwise; the entry set can only shrink. $\square$

Both directions are instances of a single monotone comparative-statics lemma, formalised in Lean as `entry_monotone` and `nonentry_monotone`; Lemma 1 is `single_crossing` (`lean/EntryContest.lean`). None depends on any axiom.

Note the prize effect applies to rich and poor challengers alike, which was the referee’s objection to the prior version (§8.2); it is accommodated here rather than assumed away.

6.10 P9: inequality and aggregate expenditure

Theorem 4 (P9). *Let $w \mapsto \bar{\mu} + \lambda(w - \bar{\mu})$, $\lambda > 1$, $\bar{\mu} = \mathbb{E}[w]$, be a linear mean-preserving spread. Then:*

1. *if the marginal entrant is richer than the mean, $w_{(k^*)} > \bar{\mu}$, then k^* weakly rises;*
2. *if the marginal entrant is poorer than the mean, $w_{(k^*)} < \bar{\mu}$, then k^* weakly falls.*

Proof. Order statistics map as $w_{(j)} \mapsto \bar{\mu} + \lambda(w_{(j)} - \bar{\mu})$, and

$$(\bar{\mu} + \lambda(w_{(j)} - \bar{\mu})) - w_{(j)} = (\lambda - 1)(w_{(j)} - \bar{\mu}),$$

which is positive exactly when $w_{(j)} > \bar{\mu}$. Since κ is decreasing in w , the left side of (3) falls for entrants richer than the mean and rises for those poorer. By Proposition 8 the right side $\Delta(j - 1)$ is unchanged. Applying the monotone comparative-statics lemma in each direction gives the two cases. $\square$

Formally this is P9-gen (§6.11) instantiated at the pivot $x_0 = \bar{\mu}$: `linSpread_isPivotSpread` machine-checks that $Tw = \bar{\mu} + \lambda(w - \bar{\mu})$ with $\lambda \geq 1$ satisfies the `PivotSpread` hypothesis, so the two branches follow from `entry_preserved_above_pivot` and `nonentry_preserved_below_pivot` without a separate argument. The displacement identity, its sign on each side of the pivot, and mean preservation are checked symbolically in `S15a-c`.

This closes what the previous draft listed as an open item. The earlier version proved only the partial effect at fixed Δ and flagged the congestion feedback as unresolved. Proposition 8 shows there is no feedback to resolve: Δ is a function of the entrant count alone and is invariant to the wealth distribution, so the partial effect is the full effect.

The sign therefore flips exactly at $w_{(k^*)} = \bar{\mu}$, i.e. at entry rate $1 - \Lambda(\bar{\mu})$ — a sharp, distribution-specific prediction rather than a qualitative “exclusive versus mass” statement. Verified on $\Lambda = \text{lognormal}(0.6, 0.45)$: predicted flip at entry rate 0.4110, confirmed at all seven threshold quantiles tested. Both branches are exercised separately in `checks/verify_p89.py`: 230 cases for the rise branch, 4000 for the fall branch (629 of them strict), with no violations.

Why this matters. Inequality *raises* status expenditure in exclusive competitions and *lowers* it in mass ones. Non-tautological, since V is fixed by construction, and it reconciles the motivating empirical pattern with the received result — inequality dampens status competition, as in Cole et al. [1992] (§IV.A) and, on the intensive margin, Hopkins and Kornienko [2004, 2010] — rather than contradicting it: that finding is recovered as the mass-competition (below-pivot) case, while exclusive competitions sit in the regime where the sign flips. The full statement, with precedents and scope, is §7.

6.11 P9 general: arbitrary pivot-spreads

The linear spread is a special case. Call T a *pivot-spread about x_0* if T is increasing and

$$T(w) \geq w \text{ for } w \geq x_0, \quad T(w) \leq w \text{ for } w \leq x_0.$$

Theorem 5 (P9, general form). *Let T be a pivot-spread about x_0 . If the marginal entrant satisfies $w_{(k^*)} > x_0$ then k^* weakly rises; if $w_{(k^*)} < x_0$ then k^* weakly falls.*

Proof. T increasing preserves the wealth ordering, so the j -th order statistic maps as $w_{(j)} \mapsto T(w_{(j)})$. Since κ is decreasing in w , the pivot property gives $\kappa(T(w_{(j)})) \leq \kappa(w_{(j)})$ for $w_{(j)} \geq x_0$ and the reverse for $w_{(j)} \leq x_0$. By Proposition 8 the gain schedule $\Delta(j-1)$ is unchanged. Apply the monotone comparative-statics lemma in each direction. $\square$

Correction to the linear statement. The relevant threshold is the *pivot* x_0 , not the mean. The two coincide for the linear spread $T(w) = \bar{\mu} + \lambda(w - \bar{\mu})$, which is why the mean appeared in §6.10; a general mean-preserving spread may pivot elsewhere, and then the flip is at x_0 . Tested directly: applying the mean-based rule to non-linear pivot-spreads produces 968 counterexamples, while the pivot-based rule has none.

The pivot-spread class is not ad hoc. It is, up to a crossing condition, the standard dispersion order of this literature. Hopkins and Kornienko [2009] introduce the *dispersive order* of Shaked [1982], defined earlier as “ordering in dispersion” by Lewis and Thompson [1981] (p. 77), precisely to sign comparative statics of a competitive equilibrium under a change in the underlying distribution — the class of problem P9-gen belongs to — and their Definition 1 says $F \leq_d G$ whenever $G^{-1}(r) - F^{-1}(r)$ is weakly increasing. Transporting this through $T = G^{-1} \circ F$ makes the displacement $T(w) - w$ weakly increasing, and then two things follow, both machine-checked in `lit/hopkins_kornienko/Dispersive.lean`: monotonicity of T is *implied* rather than assumed (`dispersive_imp_monotone`), and a zero of the displacement yields the `PivotSpread` hypothesis about that point (`dispersive_crossing_imp_pivot`). So the two hypotheses of Theorem 5 are delivered together by a standard order plus a crossing.

The crossing condition is not decoration. A pure translation $T(w) = w + 1$ is dispersive and monotone yet is a pivot-spread about *no* point at all (`shiftUp_not_pivotSpread`), so the dispersive order alone does not suffice.

Precedent. The closest structural precedent for the pivot rule is Costrell and Loury [2004]’s two-job model, where the wage schedule is governed by the single sufficient statistic $\hat{\mu} = F^{-1}(\theta)$, the ability of the marginal worker between jobs, and they state that the effect of a more unequal ability distribution “depends on whether the worker on the margin between jobs, at quantile θ , lies in the upper or lower tail”. P9-gen is the entry-count analogue. One difference should be stated rather than glossed: their marginal quantile θ is *fixed by technology*, since jobs come in fixed proportions, so only $\hat{\mu}$ moves when F does; here k^* is *endogenous* and is precisely what the comparative static is about. The analogy is in the sign logic, not in the status of the margin, and their object is a continuum wage schedule where ours is an integer count.

Their Proposition 10 is also a caution worth carrying: under Cobb–Douglas crowding, where the assignment re-optimises, the direction of the inequality effect *reverses* relative to their fixed-proportions Proposition 6. The analogue of fixed assignment here is that Δ does not move under a wealth spread — and that is not an assumption but Proposition 8. The literature’s own experience is that this sign flips when the allocation re-optimises; P9 is safe because the absence of feedback was proved, not assumed.

The hypothesis is not vacuous. Maps that contract near the pivot — for instance $T(w) = x_0 + \text{sign}(d)|d|^p$ with $d = w - x_0$ and $p > 1$, for which $|d|^p < |d|$ when $|d| < 1$ — are increasing but are *not* pivot-spreads, and they do violate the sign rule (729 violations in the control run). Four qualifying families (linear, quadratic, mixed, exponential) give zero violations across 1500 economies each. Verified in checks/verify_p9gen.py; the two cost lemmas are `cost_falls_above_pivot` and `cost_rises_below_pivot`, with order preservation as `order_preserved` (lean/EntryContest.lean, no axioms).

6.11.1 N1: the margin is endogenous, and the rule survives it

Theorem 5 is stated at the marginal index. That index is not fixed: k^* is determined by (3) and moves when the distribution moves, so the marginal entrant is in general a different agent before and after the spread. The rule nevertheless needs only a single comparison at the *pre-spread* margin.

Proposition 9 (endogenous margin). *Let T be a monotone pivot-spread about x_0 , let challengers be sorted descending, and let k^* be the pre-spread entrant count.*

1. *If $w_{(k^*)} > x_0$ then no pre-spread entrant exits, so the post-spread count satisfies $k^{*'} \geq k^*$.*
2. *If $w_{(k^*)} < x_0$ then no pre-spread outsider — that is, no agent at a rank $j > k^*$ — enters, so $k^{*'} \leq k^*$.*

No post-spread margin need be located.

Proof. Descending order does the work. For (i), any $j \leq k^*$ has $w_{(j)} \geq w_{(k^*)} > x_0$, so every inframarginal entrant is also above the pivot; by the pivot property $T(w_{(j)}) \geq w_{(j)}$, so κ falls weakly and (3) continues to hold at every such j . For (ii), any $j > k^*$ has $w_{(j)} \leq w_{(k^*)} < x_0$, so every *outsider* is also below the pivot, $T(w_{(j)}) \leq w_{(j)}$, κ rises weakly, and (3), which failed at those j by the definition of k^* , continues to fail. The argument is confined to $j > k^*$: at the margin itself (3) holds, so there is nothing there to preserve. Monotonicity of T preserves the descending order, so Lemma 1 applies to the post-spread profile and $k^{*'}$ is well defined; Δ is unchanged by Proposition 8. $\square$

This is stronger than the fixed-index reading of the same rule, and it is the respect in which the present setting differs from its closest structural precedent. In Costrell and Loury [2004]’s two-job model the marginal quantile θ is fixed by technology — jobs come in fixed proportions, so θ does not move when the ability distribution does, and only $\hat{\mu} = F^{-1}(\theta)$ shifts. Here the marginal index is itself the object the comparative static is about, and the sign rule holds anyway.

That difference should not be read as this proposition dominating theirs, and it is not claimed here. Costrell–Loury sign a smooth aggregate — a wage schedule — under a general spread, which is a harder

object than an integer threshold count and is why their argument needs machinery this one does not. What is being compared is not two theorems of differing strength but two formulations: moving the choice to the extensive margin replaces the aggregate with a threshold statistic, and it is that replacement which makes an endogenous margin tractable at all. The modelling choice is the contribution; the proposition is what the choice buys, and it is correspondingly easy once the choice is made.

The hypothesis is not slack: comparing the pivot against the *richest* challenger rather than the marginal one produces violations in 154 of 2744 control cases (checks/verify_p9gen.py, N1 CONTROL), so the marginal index is the load-bearing comparison point. Machine-checked as `entry_preserved_below_marginal_index` and `nonentry_preserved_beyond_marginal_index`, with `descending_preserved` for well-definedness; checked numerically with k^* recomputed on both sides over 20000 economies, 0 violations, the count actually moving in 393 of them.

6.11.2 R1: the margin condition, and the spreads it covers

Theorem 5 assumes T is a pivot-spread, so the displacement changes sign at most once, which is the dispersive case. A general mean-preserving spread need not be dispersive, since its quantile difference may cross zero any number of times. The pivot hypothesis is stronger than the argument requires, and so is the rank-by-rank condition an earlier version of this section stated.

Compare the two *sorted* profiles rank by rank. The rank-wise comparison of two distributions is their quantile coupling, $w'_{(j)} - w_{(j)} = \Lambda_\beta^{-1}(r_j) - \Lambda_\alpha^{-1}(r_j)$, so no assumption is needed to obtain it.

Proposition 10 (margin condition). *Let w and w' be two wealth profiles, each sorted descending, and let k^* and k^{*f} be the entry counts they induce through (3), a challenger whose wealth is at or below c not entering.*

- (i) *If $w'_{(k^*)} \geq w_{(k^*)}$, then $k^{*f} \geq k^*$.*
- (ii) *If $w'_{(k^*+1)} \leq w_{(k^*+1)}$, then $k^{*f} \leq k^*$.*

Nothing is assumed about any rank other than the marginal entrant in (i) and the first outsider in (ii).

Proof. (i) The marginal entrant has $c < w_{(k^*)} \leq w'_{(k^*)}$, so $\kappa(w'_{(k^*)}) \leq \kappa(w_{(k^*)}) \leq \Delta(k^* - 1)$. For $j \leq k^*$, sorting gives $w'_{(j)} \geq w_{(j)} > c$, so $\kappa(w'_{(j)}) \leq \kappa(w'_{(k^*)}) \leq \Delta(k^* - 1) \leq \Delta(j - 1)$, and every rank up to k^* enters after the change. (ii) If $k^* = Q$, then $k^{*f} \leq Q = k^*$. Otherwise rank $k^* + 1$ fails before the change. If $w'_{(k^*+1)} > c$, then $c < w'_{(k^*+1)} \leq w_{(k^*+1)}$ and $\kappa(w'_{(k^*+1)}) \geq \kappa(w_{(k^*+1)}) > \Delta(k^*)$, so rank $k^* + 1$ fails after the change too, and by the prefix property no rank beyond it enters. $\square$

The reduction to two ranks comes from the prefix structure of the new profile. Entry at the marginal entrant's rank carries to every richer rank, and failure at the first outsider's rank caps the count. The rank-by-rank condition of the earlier version, $w'_{(j)} \geq w_{(j)}$ for every $j \leq k^*$ or $w'_{(j)} \leq w_{(j)}$ for every $j > k^*$, is a special case, and so are Theorem 5 and Proposition 9, with the marginal entrant at or above the pivot in the rise branch and at or below it in the fall branch. Machine-checked as `margin_rise` and `margin_fall`, with `tail_rise` and `tail_fall` for the rank-by-rank form and `pivot_rise` and `pivot_fall` for pivot-spreads (lean/mathlib/Spreads.lean). Each statement quantifies over arbitrary profiles w and w' , so the displacement may cross zero any number of times away from the two ranks the statement reads.

The hypotheses are worth reading for what they omit. Branch (i) uses three properties and no others. κ is non-increasing in wealth above c , which is burden-monotonicity (§4), Δ is non-increasing (Corollary 6.3.1), and the new profile is sorted. Branch (ii) uses the first property alone. There is no requirement that w' be obtained from w by a spread or by any transformation at all, and $\sum_j (w'_{(j)} - w_{(j)}) = 0$ is nowhere used. The proposition is therefore not a comparative-static result about Λ but a monotonicity result about a threshold statistic, which is then *used* to obtain comparative statics whenever a change of profile satisfies its condition.

The pivot cannot be replaced by the mean. Take $c = 1$, CRRA utility with $\gamma = 2$, so that $\kappa(w) = 1/(w-1) - 1/w$, and the gains $\Delta(m) = 1/(m+2)$ of the benchmark of §6.5 with $V = 1$. The profiles (2.9, 2.8, 2.0) and (3.6, 2.2, 1.9) have the same total, and the second majorizes the first. The marginal entrant lies above the mean before the change, and the count falls from 2 to 1 (`mps_lowers_count`). The marginal entrant's wealth falls, so branch (i) does not apply. The first outsider's wealth falls too, so branch (ii) applies, and branch (ii) allows the fall. An earlier version of this paragraph rested the point on sampling (`checks/verify_p9gen.py`, R1-SCOPE), which remains as an illustration.

The assignment literature needs more machinery than this for a reason. The entry count k^* is a threshold statistic, and the aggregates of the assignment literature are integrals. Suen [2007] and Costrell and Loury [2004] sign smooth aggregates and must therefore control the quantile difference at every rank, which is what forces integration by parts and a log-concavity or monotone-weight condition. The entry count reads the quantile difference at two ranks, the marginal entrant and the first outsider, so no condition on the rest of the quantile function is needed.

6.11.3 The band: which margins the margin condition covers

Write the rank-by-rank displacement of the two sorted profiles as $d_j = w'_{(j)} - w_{(j)}$, rank 1 richest. The margin condition settles the direction of k^* unless

$$d_{k^*} < 0 < d_{k^*+1},$$

that is, unless the marginal entrant becomes poorer while the first outsider becomes richer (`branch_of_not_band`). A margin with this property lies in the *band*. In the Lean development the ranks are counted from 0, so the band reads $d(k-1) < 0 < d(k)$ there (`InBand`).

Dispersive changes leave no margin in the band. If $d_j > 0$ implies $d_i \geq 0$ for every $i < j$, which is single crossing from positive to negative down the ranks, no margin lies in the band (`not_band_of_single_crossing`). A pivot-spread has this property, since $d_j > 0$ forces $w_{(j)} > x_0$ and then $w_{(i)} \geq w_{(j)} > x_0$ for $i < j$ (`pivot_single_crossing`). So the rule holds at every margin for every pivot-spread.

Inside the band the signs settle nothing, for mean-preserving spreads too. With $Q = 4$, the same κ and the same benchmark gains, the profile (4, 2.8, 2.2, 2) has $k^* = 2$. The profiles (4.1, 2.7, 2.6, 1.6) and (4.6, 2.25, 2.25, 1.9) have the same total as the original profile, and each majorizes it. Both displacements have the signs (+, -, +, -) at every rank, both put the margin in the band, and the first moves the count to 3 while the second moves the count to 1 (`mps_band_signs`, `mps_band_both_directions`). The three counts hold for every gain schedule in a box around the benchmark gains, and the benchmark gains lie strictly inside that box (`mps_witness_counts`). The model's gains converge to the benchmark gains as $\mu \rightarrow 0$, so the same counts hold for every small $\mu > 0$ with r and s uniform and an investing incumbent (`mps_band_both_directions_positive`). No rule that uses only the signs of the displacement can therefore sign the change inside the band, and mean-preservation does not keep a margin out of the band. A second pair with $Q = 3$, whose changes do not preserve the total, gives the same conclusion with the signs (0, -, +) (`band_both_directions`, `band_both_directions_positive`).

The earlier band. An earlier version of this section defined the band through L , the richest rank whose wealth falls, and M , the poorest rank whose wealth rises, and reported sampled frequencies for it (`checks/verify_tailband.py`, T1-T5). That band belongs to the rank-by-rank condition and is superseded, and so are the frequencies. The coverage criterion of the earlier band remains machine-checked

as `lean/EntryContest.lean::band_empty_iff`. The sentence of that version saying the rank-by-rank condition was the whole of what the sign pattern supports was wrong, since the margin condition is weaker and still settles the direction.

What remains open. One question remains open. Some statistic of the change other than its signs might predict the direction of k^* inside the band, for instance its magnitudes or a weighted functional of the quantile difference of the kind Costrell and Loury [2004] use. The witnesses above say that the signs do not, and the witnesses say nothing about whether another statistic might.

6.12 P9 strictness

Proposition 11. *Suppose the marginal entrant is strictly below the pivot, $w_{(k^*)} < x_0$, and consider the linear family $T_\lambda(w) = x_0 + \lambda(w - x_0)$. Then $k^*(\lambda)$ is non-increasing in λ , and there exists $\bar{\lambda}$ such that $k^*(\lambda) < k^*(1)$ for all $\lambda > \bar{\lambda}$.*

Proof. Non-increasingness is Theorem 5 applied at each λ . For strictness: κ is continuous and strictly decreasing in w , with $\kappa(w) \rightarrow \infty$ as $w \downarrow c$. Since $w_{(k^*)} < x_0$, $T_\lambda(w_{(k^*)})$ decreases without bound in λ , so $\kappa(T_\lambda(w_{(k^*)}))$ eventually exceeds the fixed value $\Delta(k^* - 1)$ and the marginal entrant exits. Because the entry set is a prefix (Lemma 1), losing the marginal entrant strictly reduces the count. $\square$

The three steps are separated in the verification. That $T_\lambda(w_{(k^*)})$ is strictly decreasing in λ and unbounded below is machine-checked (`linSpread_strict_anti_in_lam`, `linSpread_unbounded_below`). That losing the marginal entrant strictly reduces the count — because the post-spread entry set is a prefix, so no index at or above the exiting one can enter — is machine-checked (`entry_ceiling_of_marginal_exit`, `strict_drop_of_marginal_exit`). The remaining step is analytic and is *not* proved in Lean: that κ diverges at the support floor, so that a low enough wealth makes the entry cost exceed the fixed gain $\Delta(k^* - 1)$. It enters as the boundary condition stated in Section 4 and is checked symbolically for the log and CRRA families in `checks/verify_sympy.py` (S16, S17). Verified in addition over 400 qualifying economies: k^* is monotone in λ and strictly lower at $\lambda = 100$ than at $\lambda = 1$ in every case.

6.13 P-MU: the Q comparative static, characterised

Proposition 12. *For $Q \geq 1$, $\Delta(0, Q+1) - \Delta(0, Q) = -V \int W (dF - dG)$ with $W \equiv CG^{Q-1}(1 - G)$, where C is the incumbent's CDF. With $V > 0$, $\Delta(0)$ rises from Q to $Q+1$ if and only if*

$$\int W dF < \int W dG.$$

The identity holds for every incumbent CDF C (`lean/mathlib/PMU.lean`, `pmu_identity`, `pmu_sign_iff`). The factor $G^{Q-1}(1 - G)$ rises in G up to $(Q-1)/Q$ and falls beyond it (`kernel_rises`, `kernel_falls`). First-order stochastic dominance alone does not sign the comparison. Two pairs with $F \leq G$, no atoms and an investing incumbent give opposite signs at $Q = 1$ and at $Q = 2$. The pair $F = x$, $G = 1 - (1 - x)^2$ on $[0, 1]$ gives $V/60$ and $V/420$, while the pair $F = x^2$, $G = x$ gives $-VQ/((Q+2)(Q+3)(Q+4))$ at every Q (`lean/mathlib/PMUwitness.lean`, `fosd_does_not_sign`, `pairB_step`).

Provenance of the weight. The factor $G^{Q-1}(1 - G)$ has the form of the log-supermodular kernel $F^{k-1}(1 - F)$ that Ryvkin and Drugov [2020] use for their comparative statics in the number of players, under $G \leftrightarrow F$ and $Q \leftrightarrow k$. The peak at $(Q-1)/Q$ is our computation, which Ryvkin and Drugov [2020] do not state. The extra factor C here is the incumbent's CDF and carries no Q . Their corresponding object is $b_k = \mathbb{E}[f(X_{(k-1:k-1)})]$, the individual marginal benefit of effort, which is the right comparator for $\Delta(0)$, whereas their hazard-rate result concerns aggregate effort and a different order statistic.

The single-crossing step carries over. Since $f - g = -\varphi'$, the identity reads $\Delta(0, Q+1) - \Delta(0, Q) = V \int G^{Q-1}(1 - G)C\varphi' dx$, and $C\varphi'$ crosses $+-$ whenever φ is single-peaked, the orientation that the variation-diminishing step of Ryvkin and Drugov [2020] needs. An earlier version of this paragraph dropped the leading minus sign of the identity and concluded that the orientation was reversed. The step is proved on measures, without densities. If $dF - dG$ is nonpositive on $(-\infty, x_0]$ and nonnegative on (x_0, ∞) , then for $1 \leq Q_1 \leq Q_2$

$$\Delta(0, Q_2+1) - \Delta(0, Q_2) \leq G(x_0)^{Q_2-Q_1} [\Delta(0, Q_1+1) - \Delta(0, Q_1)]$$

(pmu_single_crossing). Once the difference is nonpositive it stays nonpositive, so $\Delta(0, \cdot)$ is quasi-concave on $Q \geq 1$, rising and then falling, where either phase may be absent (pmu_quasiconcave). The hypothesis holds at $x_0 = \mu$ when r is uniform on $[0, 1]$, for any law of $s \geq 0$ and any $0 < \mu \leq 1$ (uniform_left, uniform_right, pmu_quasiconcave_uniform). With r uniform the gain never falls (§6.8, Delta_mono_uniform), so the falling phase needs another law of r . It occurs for r the smaller of two uniform draws, where the gain rises from $Q = 1$ to $Q = 2$ and falls from $Q = 2$ to $Q = 3$ (lean/mathlib/FallWitness.lean, rise_then_fall). The hypothesis holds for a class of primitives that contains both cases. Let the investment lift the score by $1 - \mu$ with probability $1 - p$ and by nothing otherwise, let r put no mass at or below 0, and let the law of r never lose mass when a set above $c > 0$ moves down by c , which holds when r has a density that does not rise on the positive reals. Then for every $0 < \mu < 1$, every p and every incumbent, $dF - dG$ changes sign once, at $x_0 = 1 - \mu$, and $\Delta(0, \cdot)$ is quasi-concave in Q (lean/mathlib/ShiftClass.lean, quasiconcave_of_shiftMono, shiftMono_of_density). The density $2(1 - r)$ of the witness above lies in the class (witness_quasiconcave).

When the gain must fall. Within the class, the gain must fall, and not only may, under a strict version of the same condition. Let G_t be the non-investor's law shifted up by the lift $t = 1 - \mu$. Suppose that on some interval $(a, b]$ with $t < a < b$ the shifted law puts strictly more mass than G , that $G(t) < G(a)$, $G(b) < 1$ and $C(a) > 0$, and that $p < 1$. Then $\Delta(0, Q+1) < \Delta(0, Q)$ for every Q from some Q_0 on (lean/mathlib/MustFall.lean, must_fall). The proof bounds the part of the step below t by $G(t)^{Q-1}$ times a constant, where the shifted law has no mass, and bounds the part above a from below by $G(a)^{Q-1}$ times a positive constant, so the ratio $(G(t)/G(a))^{Q-1}$ decides the sign (must_fall_general). The base density $2(1 - r)$ meets the condition on $(a, b] = (1/2, 5/8]$ for every $p < 1$ (witness_must_fall), while a uniform base law meets it on no interval (uniform_fails_gain).

7 Contribution and its precedents

The claim. For a fixed prize awarded by rank, the effect of resource inequality on *how many people compete* has a determinate sign, and the sign is set by a single observable: whether the marginal entrant stands above or below the pivot of the spread. A *dispersive* spread of the resource distribution — one whose displacement changes sign exactly once, the leading case being the linear mean-preserving spread $w \mapsto \bar{\mu} + \lambda(w - \bar{\mu})$ of Theorem 4 — *widens* entry when the marginal competitor is richer than the pivot and *shrinks* it when poorer (Theorem 5), strictly so under the boundary condition of Proposition 11. What the proof uses is weaker still. The rise branch reads the quantile difference at the marginal entrant alone, the fall branch reads the quantile difference at the first outsider alone, and mean-preservation is nowhere among the hypotheses (Proposition 10). Within that class the rule holds globally, distribution-free, and under burden-monotonicity of u alone ((2); strict concavity is sufficient but not necessary, Claim 1).

That class is a genuine restriction, and it belongs with the claim rather than only with the scope discussion at the end of this section. A Rothschild–Stiglitz mean-preserving spread need *not* be dispersive, since its quantile difference may cross zero any number of times. With the marginal entrant above the mean, a change that preserves the total and majorizes the original profile can lower k^* (§6.11.2, mps_lowers_count), so the

rule stated with the mean in place of the pivot fails for such changes. The sign rule is therefore a theorem about dispersive spreads and about the margin condition, and not about mean-preserving spreads as such, and the rule should not be quoted in the latter form. Which changes the margin condition covers is settled in §6.11.3. The condition is silent exactly when the marginal entrant becomes poorer and the first outsider becomes richer. Dispersive changes never reach that band, and inside the band the displacement signs determine nothing, for mean-preserving spreads as for other changes.

The economic content is a reconciliation. The received result — inequality dampens status competition — goes back to Cole et al. [1992], whose §IV.A example has the more compact income distribution producing the higher savings rate, and is developed on the intensive margin by Hopkins and Kornienko [2004] and Hopkins and Kornienko [2010], where less dispersed endowments intensify conspicuous consumption. That finding is here recovered as one branch: mass competitions, in which the marginal entrant sits low in the distribution. Exclusive competitions, in which the marginal entrant is already wealthy, sit on the other branch, where inequality *feeds* entry. The received result is the below-pivot case of a two-branch rule, not the general case. This is non-tautological because V is fixed by construction (§6.9): nothing about the prize moves when the distribution spreads.

The margin is endogenous and the rule survives that (Proposition 9): the hypothesis is a single comparison at the pre-spread margin, and the wealth ordering carries the rest, so no post-spread margin need be located. Costrell and Loury [2004], the closest structural precedent, work with a marginal quantile fixed by technology. The claim here is not that the present result is the stronger one — they sign a smooth aggregate under a general spread, which is the harder object — but that the extensive-margin formulation replaces that aggregate with a threshold statistic, and an endogenous margin is tractable in the second setting and not the first. The formulation is what is being offered.

Where several entrant sets are equilibria, the assortative one is the cheapest, and under an anonymous gain schedule it therefore maximises the unweighted sum of payoffs (Proposition 6) — so using it is a stated selection with an argument behind it, not an assumption. That argument is not a welfare claim, for the reasons set out with the proposition.

Two features make the claim well-posed, and both required proof. The object being signed is an invariant of the game: every pure-strategy equilibrium has the same entrant count (Proposition 6.4.1), so “how many compete” does not depend on which equilibrium is selected — even though *who* competes does (§4, Information). And the class of spreads for which the rule is proved is not bespoke: a pivot-spread is exactly a dispersive-order shift with a crossing point (§5), the dispersive order being the inequality order this literature already uses [Hopkins and Kornienko, 2009, Shaked, 1982].

Why the extensive margin is the point. The intensive-margin version of the same question exists and is *not* sign-determinate. Schroyen and Treich [2016] sign a mean-preserving spread in wealth inside the same privilege-contest payoff: their Theorem 3 gives the effect as positive iff $2A(1 - m^2) > P$, with $A = -u''/u'$ and $P = -u'''/u''$ — local, two players, and conditioned on the contest technology and the *third* derivative of utility. The dependence on u''' is not cosmetic: at $m = 1/2$, CARA with $a = 1$ and log utility at $w = 1$ share $A = 1$ but have $P = 1$ and $P = 2$, and give opposite signs. The extensive-margin rule of Theorem 5 needs only that κ is strictly decreasing, which is $u'' < 0$. So the margin is not a modelling convenience; it is what makes inequality’s effect on competition determinate at all. An anticipated objection should be met here rather than left to a referee: the two results answer the same economic question — what does a spread of the resource distribution do to competitive expenditure — in different models, and the comparison of hypotheses across margins is legitimate precisely because the margin is the object of choice being defended. It is also not robust terrain for the intensive side: in the assignment literature the direction of the inequality effect reverses when the allocation re-optimises (Costrell and Loury [2004], Proposition 10 against their Proposition 6), so conditioning results of this kind on the margin is already the literature’s own practice.

The 1981 precedent. Lazear and Rosen [1981], in the “Income Distributions” subsection of their §III, already have agents self-selecting into a tournament by endowed wealth. This must be acknowledged, with two qualifications that bound what it concedes.

First, their claim is offered as an example and they say so: the subsection opens “While it is not possible to make a general argument based on an example, table 1 suggests...”, and rests on one utility function, quadratic costs, normal errors and two endowment levels. They *suggest* the sorting under declining absolute risk aversion; the present model *derives* it, distribution-free.

Second, the channel differs, and decisively. Their sorting runs through DARA, which is a restriction on u''' : $A'(w) < 0$ has numerator $(u'')^2 - u'''u'$ and so requires $u''' > 0$. Here the sorting runs through $\kappa(w) = u(w) - u(w - c)$ being decreasing, which follows from $u'' < 0$ alone. Quadratic utility separates them: it is concave, has $u''' = 0$, and exhibits *increasing* absolute risk aversion — so κ still sorts while DARA fails outright.

Third, and least often noticed, the object of choice differs. Theirs is a choice between two compensation schemes that both pay, so the sorting is tolerance for the binomial spread of a lottery against a concentrated alternative. Ours is whether to pay a fixed entry cost at all, so the sorting is affordability in utility units. That is a difference in the economics, not only in a derivative.

What is not claimed. The mechanism — wealth sorting entry through concavity — is not novel; the two precedents above own it between them, and the sentence claimed here is the two-branch sign rule, not the sorting. The remaining results are apparatus in its service and are presented as such. The uniform cap of §6.7 is the deterministic, heterogeneous, utility-unit form of an accounting bound already present in Fu et al. [2015] (equation (2) with Definition 2) and Fu and Lu [2010] (equation (7)); it is used here to dispose of the ceiling objection, not claimed as new. The characterisation of §6.13 runs on the order-statistic kernel of Ryvkin and Drugov [2020], whose message — that no competitor-number prediction is universal across noise distributions — is theirs in print; what is added is the extensive-margin instance that refutes the referee’s conjecture. And the scope of the central claim is stated with it. The rule is proved under the margin condition of Proposition 10, which reads the quantile difference at the marginal entrant for the rise branch and at the first outsider for the fall branch. Pivot-spreads are the special case in which one branch applies at every margin. A mean-preserving spread is covered when the displacement is non-negative at the marginal entrant or non-positive at the first outsider, and the displacement may cross zero freely at the other ranks. The condition is a restriction and not a formality, since two mean-preserving spreads with the same displacement signs at every rank can put the margin in the band and move k^* in opposite directions (§6.11.3). What is not settled is whether some statistic of the change other than its signs predicts the direction inside the band (§9, item 3).

8 How this addresses the referee reports

8.1 Asymmetric k -of- Q equilibria

Both referees identified the same defect: the prior version compared aggregate participation regimes rather than checking unilateral deviations, which is not Nash logic for a finite-player game. Proposition 6.4 (§6.4) replaces the binary all-in/all-out structure with the asymmetric k -of- Q equilibrium requested, derived from first principles rather than assumed.

8.2 Prize versus affordability

Referee 2 objected that, with $y_2 - y_1$ serving simultaneously as “inequality” and the contest prize, the claim that inequality raises status expenditure was close to tautological, and that a rise in the prize should raise *all* participants’ incentives, not only the incumbent’s. Decoupling V from the wealth distribution (§4) and

separating the two channels in Proposition 6.9 (§6.9) addresses this directly; Theorem 4 (§6.10) then states the inequality result as a genuine theorem rather than a definitional artefact.

8.3 The mobility ceiling

Both referees identified the old ceiling as an artefact: mechanical arithmetic under uniform supports, a $Q \rightarrow \infty$ limit inconsistent with the equilibrium existing, and a fixed prize with unbounded entry not corresponding to anything. Proposition 6.7 (§6.7) replaces it with the saturation bound $k^* \leq V/\kappa$, uniform in Q and requiring no limit to be taken.

9 Open items

Closed since the previous draft: the P9 fixed-point effect (there is no feedback to resolve — Proposition 8); uniqueness of the wealth fixed point (Lemma 1); the exact sign condition for the Q comparative static (§6.13); P8, now proved rather than verified; non-linear spreads (Theorem 5); and P9 strictness (Proposition 11). Also closed: the literature position, which was previously listed here as gating further proof effort. It is settled in the accompanying literature survey and stated in Section 7; the verdict is that the mechanism has precedents and the contribution is the combination of P7, P9-gen and P-MU.

What genuinely remains:

1. **The incumbent’s own entry decision** is exogenous throughout (C fixed in $H_m = F^m G^{Q-1-m} C$). Endogenising it, and testing whether an equilibrium exists in which the incumbent abstains while a challenger invests, is untouched. This is a modelling extension rather than a gap in what is claimed.
2. **Spreads that are not pivot-spreads — resolved, see Proposition 6.11.2.** This item previously read that the sign rule was open for general Rothschild–Stiglitz spreads, and proposed importing the quantile-decomposition machinery of Costrell and Loury [2004] and Suen [2007]. That turned out to be unnecessary. Those techniques sign a smooth *aggregate* under a general spread and therefore need integration by parts and, in Suen’s case, log-concavity of $1 - F$. But k^* is not an aggregate: it is a threshold statistic that reads the quantile function at the marginal entrant and the first outsider. The recorded “count versus aggregate” obstacle was real, and it runs the opposite way — the count is *easier*. What remains genuinely open is narrower.
3. **When does the margin condition hold? Resolved, with a correction.** This item previously asked for a characterisation of the mean-preserving spreads satisfying Proposition 10, and an earlier answer called the rank-by-rank condition sharp. The rank-by-rank condition was not sharp, since the margin condition reads only two ranks and still settles the direction. The margin condition is silent exactly on the band $d_{k^*} < 0 < d_{k^*+1}$, and inside the band the displacement signs determine nothing, for mean-preserving spreads as for other changes. Both facts are established in §6.11.3 and `lean/mathlib/Spreads.lean`. What remains open is narrower. Some statistic of the change other than its signs might predict the direction inside the band, and the Costrell and Loury [2004] monotone-weight technique would belong there if an aggregate version were wanted later.
4. **Strictness away from the linear family.** Proposition 11 is stated for T_λ with $\lambda \rightarrow \infty$. A characterisation of exactly which spreads move k^* strictly, rather than an asymptotic sufficient condition, is not attempted.
5. **The competitor-number effect, sharpened.** $\Delta(0, \cdot)$ is quasi-concave in Q when φ is single-peaked, and φ is single-peaked when r is uniform and, with a single-step investment, when r has a density that does not rise, and the gain must eventually fall when the lift strictly adds mass on an interval of the

non-investor's range (§6.13). Open are the location of the peak in Q and laws of s beyond a single step. An earlier version of this item conjectured an interior *minimum*, from the reversed orientation that §6.13 corrects. Under the single-peaked hypothesis an interior minimum is impossible.

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